

● HARD

● INFORMATION AND IDEAS

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09

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

The following text is adapted from Lewis Carroll's 1889 satirical novel *Sylvie and Bruno*. A crowd has gathered outside a room belonging to the Warden, an official who reports to the Lord Chancellor.

One man, who was more excited than the rest, flung his hat high into the air, and shouted (as well as I could make out) "Who roar for the Sub-Warden?" Everybody roared, but whether it was for the Sub-Warden, or not, did not clearly appear: some were shouting "Bread!" and some "Taxes!", but no one seemed to know what it was they really wanted.

All this I saw from the open window of the Warden's breakfast-saloon, looking across the shoulder of the Lord Chancellor.

"What can it all mean?" he kept repeating to himself. "I never heard such shouting before—and at this time of the morning, too! And with such unanimity!"

Based on the text, how does the Lord Chancellor respond to the crowd?

- A. He asks about the meaning of the crowd's shouting, even though he claims to know what the crowd wants.
- B. He indicates a desire to speak to the crowd, even though the crowd has asked to speak to the Sub-Warden.
- C. He expresses sympathy for the crowd's demands, even though the crowd's shouting annoys him.
- D. He describes the crowd as being united, even though the crowd clearly appears otherwise.

Several studies of sediment (e.g., dirt, pieces of rock, etc.) in streams have shown an inverse correlation between sediment grain size and downstream distance from the primary sediment source, suggesting that stream length has a sorting effect on sediment. In a study of sediment sampled at more than a dozen sites in Alpine streams, however, geologists Camille Litty and Fritz Schlunegger found that cross-site variations in grain size were not associated with differences in downstream distance, though they did not conclude that downstream distance is irrelevant to grain size. Rather, they concluded that sediment influx in these streams may have been sufficiently spatially diffuse to prevent the typical sorting effect from being observed.

Which finding about the streams in the study, if true, would most directly support Litty and Schlunegger's conclusion?

- A. The streams regularly experience portions of their banks collapsing into the water at multiple points upstream of the sampling sites.
- B. The streams contain several types of sediment that are not typically found in streams where the sorting effect has been demonstrated.
- C. The streams mostly originate from the same source, but their lengths vary considerably due to the different courses they take.
- D. The streams are fed by multiple tributaries that carry significant volumes of sediment and that enter the streams downstream of the sampling sites.

“Gestures” in painting are typically thought of as bold, expressive brushstrokes. In the 1970s, American painter Jack Whitten built a 12-foot (3.7-meter) tool he named the “developer” to apply paint to an entire canvas in one motion, resulting in his series of “slab” paintings from that decade. Whitten described this process as making an entire painting in “one gesture,” signaling a clear departure from the prevalence of gestures in his work from the 1960s. Some art historians claim this shift represents “removing gesture” from the process. Therefore, regardless of whether using the developer constitutes a gesture, both Whitten and these art historians likely agree that _____ blank

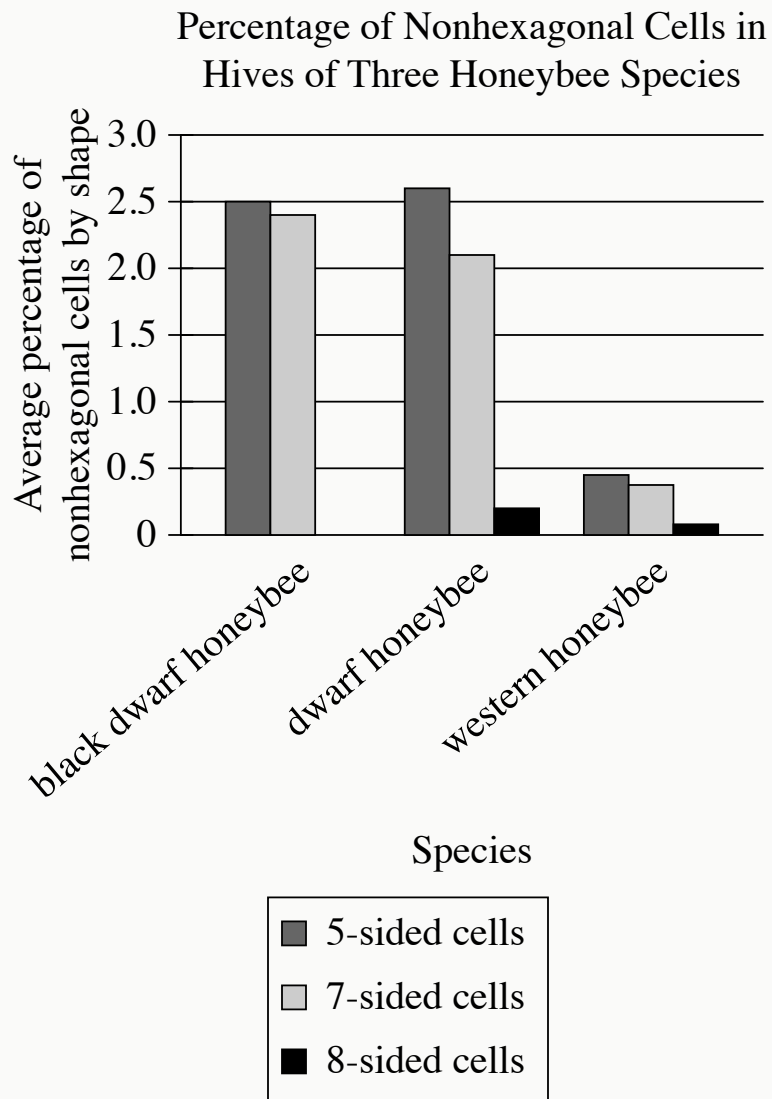
Which choice most logically completes the text?

- A. any tool that a painter uses to create an artwork is capable of creating gestures.
- B. Whitten’s work from the 1960s exhibits many more gestures than his work from the 1970s does.
- C. Whitten became less interested in exploring the role of gesture in his work as his career progressed.
- D. Whitten’s work from the 1960s is much more realistic than his work from the 1970s is.

The following text is adapted from William Shakespeare's 1609 poem "Sonnet 27." The poem is addressed to a close friend as if he were physically present.

Weary with toil, I [hurry] to my bed,
The dear repose for limbs with travel tired;
But then begins a journey in my head
To work my mind, when body's work's expired:
For then my thoughts—from far where I abide—
[Begin] a zealous pilgrimage to thee,
And keep my drooping eyelids open wide,
What is the main idea of the text?

- A. The speaker is asleep and dreaming about traveling to see the friend.
- B. The speaker is planning an upcoming trip to the friend's house.
- C. The speaker is too fatigued to continue a discussion with the friend.
- D. The speaker is thinking about the friend instead of immediately falling asleep.



- All values are approximate.
- For each data category, the following bars are shown:
 - 5-sided cells
 - 7-sided cells
 - 8-sided cells
- The average percentage of nonhexagonal cells by shape data for the 3 categories are as follows:
 - black dwarf honeybee:

- 5-sided cells: 2.5 percent
- 7-sided cells: 2.4 percent
- 8-sided cells: 0.0 percent
- dwarf honeybee:
 - 5-sided cells: 2.6 percent
 - 7-sided cells: 2.1 percent
 - 8-sided cells: 0.2 percent
- western honeybee:
 - 5-sided cells: 0.45 percent
 - 7-sided cells: 0.38 percent
 - 8-sided cells: 0.08 percent

Honeybee hives consist mainly of hexagonal (six-sided) units called cells, in which queens lay eggs. Hexagonal cells for eggs that develop into nonreproductive workers are smaller than those for eggs that develop into reproductive drones, though the size difference varies by species. Difference in cell size results in a construction problem—it's hard to neatly connect sections of small cells to sections of large cells—that worsens as the difference increases. To fill in gaps between the sections when building a hive, bees rely on cells that have more or fewer than six sides. A student studying beehive structure consults data on three species, concluding that _____ blank

Which choice most effectively uses data from the graph to complete the student's conclusion?

- A. cells for worker eggs are probably closer in size to cells for drone eggs in the hives of the western honeybee than in the hives of the dwarf honeybee and the black dwarf honeybee.
- B. both the western honeybee and the black dwarf honeybee probably reserve eight-sided cells for drone eggs, while the dwarf honeybee likely deposits drone eggs in seven-sided cells.
- C. the western honeybee probably relies on many more geometrical shapes when constructing cells than either the dwarf honeybee or the black dwarf honeybee does.

- D. the percentage of hexagonal cells is probably slightly lower in the hives of the western honeybee than in the hives of the dwarf honeybee and the black dwarf honeybee.

Marta Coll and colleagues' 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri's 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers' decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____ blank

Which choice most logically completes the text?

- A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's census.
- B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.
- C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.
- D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

To understand how temperature change affects microorganism-mediated cycling of soil nutrients in alpine ecosystems, Eva Kaštovská et al. collected plant-soil cores in the Tatra Mountains at elevations around 2,100 meters and transplanted them to elevations of 1,700–1,800 meters, where the mean air temperature was warmer by 2°C. Microorganism-mediated nutrient cycling was accelerated in the transplanted cores; crucially, microorganism community composition was unchanged, allowing Kaštovská et al. to attribute the acceleration to temperature-induced increases in microorganism activity.

It can most reasonably be inferred from the text that the finding about the microorganism community composition was important for which reason?

- A. It provided preliminary evidence that microorganism-mediated nutrient cycling was accelerated in the transplanted cores.
- B. It suggested that temperature-induced changes in microorganism activity may be occurring at increasingly high elevations.
- C. It ruled out a potential alternative explanation for the acceleration in microorganism-mediated nutrient cycling.
- D. It clarified that microorganism activity levels in the plant-soil cores varied depending on which microorganisms comprised the community.

Percentage Point Changes in US Federal Outlays Relative to GDP by Congressional Status

Period	Congressional status	Change in total outlays	Change in nondefense outlays	Change in defense outlays
1981–1988	divided	–0.4	–1.3	0.9
1975–1976	divided	2.7	3.0	–0.3
1977–1980	undivided	0.3	0.6	–0.3
1964–1968	undivided	1.9	1.4	0.5
1969–1974	divided	–1.8	2.1	–3.9

Economist Steve H. Hanke has shown that divided US Congresses—which occur when one party holds the majority in the House of Representatives and another holds the majority in the Senate—tend to accompany reductions in total federal outlays (spending) relative to gross domestic product (GDP), which Hanke interprets to reflect decreases in government size. Hanke calculated the percentage point change in total outlays (encompassing nondefense and defense outlays) for consecutive US Congresses. Hanke has pointed to his calculations as evidence that a divided Congress may be a “necessary but not sufficient condition” for a decrease in government size to occur.

Which choice best describes data from the table that support the underlined claim?

- A. The periods of undivided Congresses were associated with increases in nondefense outlays, whereas all the periods of divided Congresses except one were associated with reductions in defense outlays.
- B. All the periods of divided Congresses were associated with reductions in total outlays, although two periods were also associated with increases in nondefense outlays.
- C. The periods of undivided Congresses were associated with increases in total outlays, whereas all the periods of divided Congresses were associated with reductions in either nondefense outlays or defense outlays.
- D. All the periods of divided Congresses except one were associated with reductions in total outlays, whereas the periods of undivided Congresses were associated with increases in total outlays.

One recognized social norm of gift giving is that the time spent obtaining a gift will be viewed as a reflection of the gift's thoughtfulness. Marketing experts Farnoush Reshadi, Julian Givi, and Gopal Das addressed this view in their studies of norms specifically surrounding the giving of gift cards, noting that while recipients tend to view digital gift cards (which can be purchased online from anywhere and often can be redeemed online as well) as superior to physical gift cards (which sometimes must be purchased in person and may only be redeemable in person) in terms of usage, 94.8 percent of participants surveyed indicated that it is more socially acceptable to give a physical gift card to a recipient. This finding suggests that _____ blank

Which choice most logically completes the text?

- A. gift givers likely overestimate the amount of effort required to use digital gift cards and thus mistakenly assume gift recipients will view them as less desirable than physical gift cards.
- B. physical gift cards are likely preferred by gift recipients because the tangible nature of those cards offers a greater psychological sense of ownership than digital gift cards do.
- C. physical gift cards are likely less desirable to gift recipients than digital gift cards are because of the perception that physical gift cards require unnecessary effort to obtain.
- D. gift givers likely perceive digital gift cards as requiring relatively low effort to obtain and thus wrongly assume gift recipients will appreciate them less than they do physical gift cards.

In 2019, 20 previously unknown moons were confirmed to be orbiting Saturn. Three of the moons have prograde orbits (orbiting in the direction the planet spins), and the other 17 have retrograde orbits (orbiting in the opposite direction of the planet's spin). All but one of the 20 moons are thought to be remnants of bodies that orbited Saturn until they broke apart in collisions. Although the one exceptional moon orbits in the same direction as the planet's spin, its orbit is highly eccentric compared to the rest, which may suggest that it has a different origin than the other 19 moons.

Based on the text, which choice best describes the moon with the eccentric orbit?

- A. It doesn't have a retrograde orbit, but it likely has the same origin as the moons with retrograde orbits.
- B. Its orbit is so tilted with respect to the other moons' orbits that it's neither prograde nor retrograde.
- C. It has a prograde orbit that is likely the result of having collided with another body orbiting Saturn.
- D. It has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.